

BIO 005 Human Physiology. Patient File, Week 12. Fluid, Electrolyte, and Acid Base Chemistry

Physiologist of record: Neil Armitage. Section BIOL-5-D9286. Patient Walter B., 58, long haul driver. File PF-G-1113. Week dates Nov 23 to 29. Exported 2026-08-22 18:21. Week not yet closed.

This week on the ward

Sodium has been low on every panel and he is not on a diuretic that would explain it.

Prediction, locked before any result existed

Locked 2026-08-22 18:21

I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.

Order trail

Sequence at a glance. Tokens used 6 of 6.

#	Time	Order	Cost	Result
1	2026-08-22 18:21	Arterial pH	2	7.42
2	2026-08-22 18:21	Arterial carbon dioxide tension	2	33 mmHg
3	2026-08-22 18:21	Arterial oxygen tension	2	74 mmHg

1. Arterial pH | 2026-08-22 18:21 | 2 tokens

PREDICTED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
GAVE UP	Chief concern in the patient's own words. I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
RESULT	7.42 (reference 7.35 to 7.45)
CHANGED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.

2. Arterial carbon dioxide tension | 2026-08-22 18:21 | 2 tokens

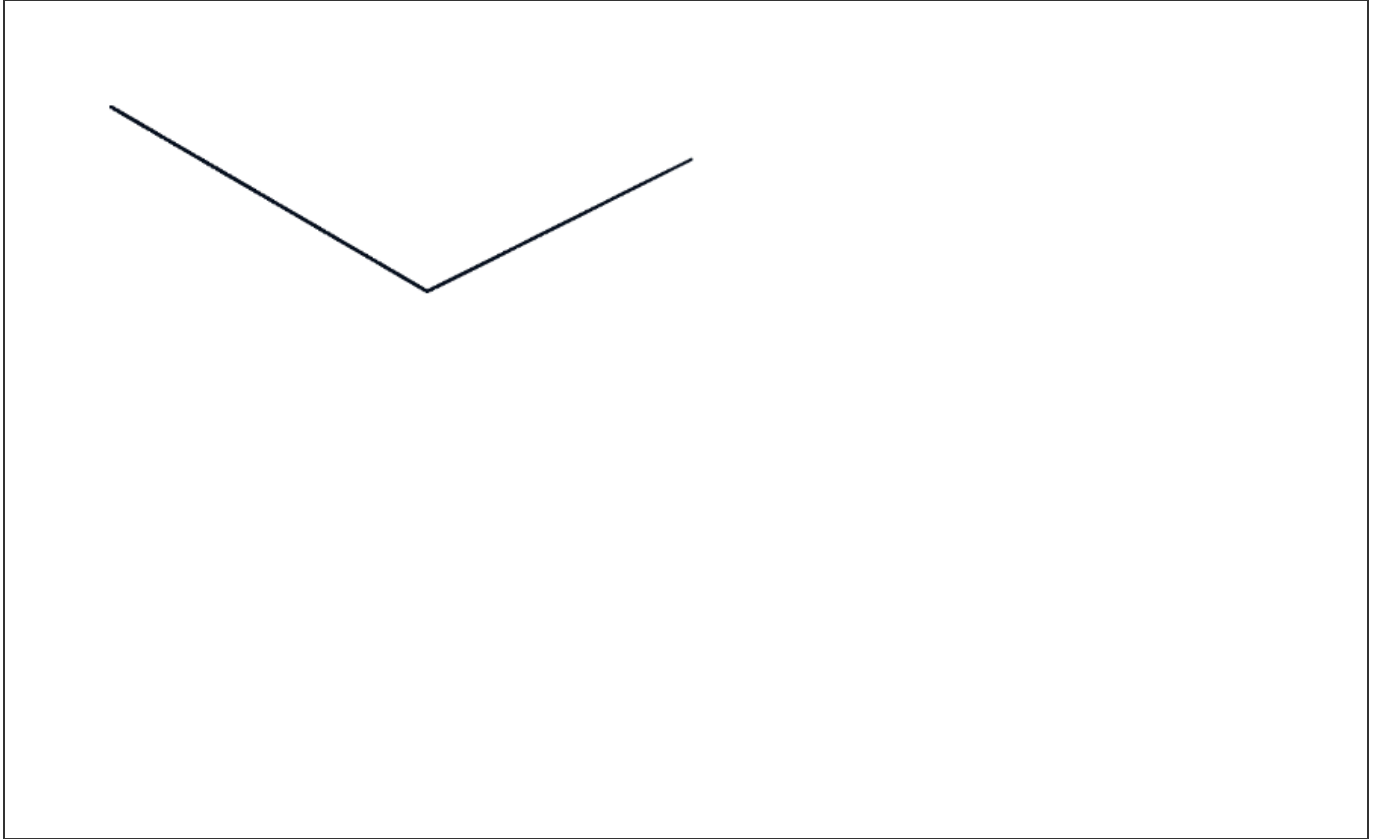
PREDICTED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
GAVE UP	Chief concern in the patient's own words. I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
RESULT	33 mmHg (reference 35 to 45 mmHg)
CHANGED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.

3. Arterial oxygen tension | 2026-08-22 18:21 | 2 tokens

PREDICTED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
GAVE UP	Chief concern in the patient's own words. I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.
RESULT	74 mmHg (reference 80 to 100 mmHg)
CHANGED	I expect this to move away from the reference range because the mechanism I believe is operating produces a predictable directional change, and the alternative explanation would predict the opposite result here.

Figure

Draw the Henderson Hasselbalch relationship as a seesaw with bicarbonate on one side and carbon dioxide on the other. Position this patient's seesaw and show the compensation arrow.



Description of the figure

Synthesis

Acid base state

Primary disturbance

Compensation

Calculate the anion gap. Show the numbers you used and state whether it is elevated.

Statement

The reasoning in this file is mine. Where I used a source, a classmate, or a generative tool to check my understanding, I have said so in the box below. I understand that the order in which I asked for information, and what I said before I saw it, is what is being graded.

Sources or tools consulted: _____

Signature and date

File fingerprint

Week 12 fingerprint 017J-QUWL. This changes if any answer on this page changes, so the copy your instructor grades is the copy you exported.